PACMAN Legacy Firmware (1.X)

Branches, build steps and bitwise reproduction

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1 Purpose and scope

PACMAN firmware comes in three orthogonal families: *legacy* (1.X, everything before consistent versioning), *hijinks* (2.X, the 2024 firmware for the 1v5 test stands, described in a companion note) and *zipline* (3.X, the current firmware in the `pacman-firmware` and `pacman-software` repositories). This note describes how the legacy firmware is kept in the repository `mulhearn/pacman`: one branch per hardware line, each holding as much of that line's firmware as possible, every state that was released or deployed reproducible from the repository alone, and the proof of each reproduction.

A bitstream is *reproduced bitwise* when the SHA-256 of its configuration body (the bitstream without its header, which carries the build date and paths) equals that of the reference bitstream. The bodies are quoted here by their first eight hex digits.

2 Layout

- One branch per hardware line (`hw1v3`, `hw1v4`, `hw1v5`), firmware only, as linear as the history allows.
- Historic commits keep their original hashes. New commits go only at a branch's tip:
  - *snapshots*: firmware that was built outside the repository, restored from its Vivado project archive;
  - *build steps*: what makes a state build from the repository alone (fixes without behavior change, staging scripts); they stay on the branch;
  - *build options*: variants of a build (rot8).
- Superseded side lines, which change nothing on the main lines, are recorded on two *excursions* branches, firmware and software.
- The layout is frozen (2026-11-08): the branches lost their `working-on-` prefix, every built state has an annotated tag, and the archive branch `legacy-archive-2026` holds all of it (Section 8). Nothing on it is rewritten.
- A checker (`check-legacy-hw1vN.sh`, run on fresh clones) enforces the layout: each spine ends at its historic tip with the original hashes and is followed by exactly the listed new commits, the build steps change only build scripts, the excursions never change the tree, and commit messages cite only commits on the spines.

The first commit of all of them is 0b9bee4 (2019-11-16, the Trenz TE0720 skeleton). Zipline shares only that commit with legacy; `pacman-firmware` and `pacman-software` share no history with this repository.

Branch	Tip	Historic tip	New at the tip
`legacy-firmware-hw1v3`	9fbc1dc	501180f	893 snapshot, build step, 883 snapshot
`legacy-firmware-hw1v4`	d08d199	21019ee	894 (two commits), build step
`legacy-firmware-hw1v5`	7062e17	2425e67	SYNC FIX build step, build option rot8
`legacy-firmware-excursions`	d5fd1c7	(2425e67)	three record merges
`legacy-software-excursions`	4a27ae1	(2425e67)	three record merges

Table 1: The legacy line branches (2026-10-31). Each is the second parent of a merge on `legacy-archive-2026`. Deleted from `mulhearn/pacman` on 2026-11-08, they are recovered from the archive (Section 8).

3 The hw1v3 line

Spine. The trunk (v1.2, v1.2b, three 2020 merges), then the v1.3 releases: v1.3-0.0 (a10a44a, 2020-12-23, the first v1.3 board), v1.3-0.1 (e26918e, external sync and trigger inputs), v1.3-0.2 (1dfc351, no firmware change), v1.3-0.3 (2099d0d, POSI/PISO invertible), and v1.3-1.0 (501180f, 2022-05-16, Peter Madigan's adjustable MCLK). Every tag builds as committed with Vivado 2018.3 (the first run fails at IP generation, the rerun builds).

New commits.

- **893** (a5f8e91): Armin Karcher's V3 firmware (2024-02), the version that runs on the 1v3 boards (version register `0x37D`), from its project archive `pacman-fw893.xpr.zip`: the archive's project Tcl as `tcl/pacman-fw15.tcl` (Vivado 2021.1) and the sources it needs that differ from 501180f (`clk_ddr_en.vhd`, `uart_en.vhd` new; `larpix_mclk_sel.vhd`, `larpix_reset_gen.vhd` changed; the FIFO `.xci` as upgraded to 2021.1).
- **Build step** (1f66dcd): `tcl/build_fw15.sh` and `tcl/build_fw15.tcl`.
- **883** (9fbc1dc): 893 with edge-sensitive SYNC/TRIG inputs (Armin, 2024-05, for the 2x2 trigger module): `larpix_reset_gen.vhd` and that archive's project Tcl.

Build. From the top of a checkout, with the Vivado 2021.1 settings sourced:

```
bash tcl/build_fw15.sh        # -> products/zsys_wrapper.bit
```

The project Tcl is Armin's export, unchanged: it names its files in the layout of his Vivado project and has no top-level wrapper. The script stages the files it requires (its own `checkRequiredFiles` list) from `srcs/`, `constraints/` and `srcs/ip` into that layout under `build/fw15`, runs the export, adds the wrapper as `tcl/build_xpr.tcl` does, and builds through `write_bitstream`.

4 The hw1v4 line

Spine. From v1.3-0.3: the rev4 commits (new constraints with one GLOBAL_SYNC, I2C virtual registers, eMMC), v1.4-0.0 (4532a23, 2022-04-12), the merge of v1.3-1.0 (b37e07f), v1.4-0.2 (0681491, 2022-10-04, the last larpix release), and Armin's module2 upload (af8c85b, 21019ee, 2023-03-01: the resethack 1.8 project Tcl `tcl/pacman_resethack1.8.tcl` and `pacman_r4.xdc`).

New commits.

- **894**, resethack 1.8 (a693c5e, c98f6b0): the sources the module2 Tcl needs that were never committed (`uart_en.vhd`, `tristate_out.vhd`, `clk_ddr_en.vhd`, `larpix_reset_gen.vhd` with the per-tile

reset mask, `zsys_wrapper.vhd`), from the project archive `pacman_resethack1.85.xpr.zip`, and the FIFO `.xci` as upgraded to 2021.1.

- **Build step** (d08d199): `tcl/build_resethack1.8.sh`, `tcl/recreate_resethack1.8.tcl`, `tcl/build_resethack1.8.tcl`.

Build. With the Vivado 2021.1 settings sourced:

```
bash tcl/build_resethack1.8.sh     # -> products/zsys_wrapper.bit
```

The recreate script stages every file Armin's Tcl requires under `build/resethack1.8`, links the Windows path by which it imports the block design (relative on Linux) to the staged tree, and sources a copy with one edit: Vivado 2021.1 places `pacman_r4.xdc` in `constrs_1/imports/constraints/`, so the target constraints path is changed to match. The build launches the implementation a second time when the first launch stops at the 2021.1 IP generation.

5 The hw1v5 line

Spine. From v1.4-0.0: the 2024 work (c2872b5, new constraints, IP upgrade, Vivado 2020.2; 36e32af, SYNC FIX: RESETN replaced by a per-tile SYNC; 5588b58, constraints cleanup; d8790e0, rot8 constraints as a separate file), ending at 2425e67 (2024-08-11). 2425e67 does not build as committed.

New commits.

- **Build step** (3155d4a, "restore build from repo"): four fixes without behavior change: `build_xpr` writes the hardware description with `write_hwdef` (Vivado 2020.2 writes no `.sysdef`); the data_tx subset converter's options are set after `validate_bd_design` (AR 58038); `larpix_mclk_sel` has integer generics, so the block design regenerates; `larpix_clk`'s `C_RW_REG3_DEFAULT` is `0x00000000`, the value always built (the source's `0X00000004` was parsed as 0). The hijinks firmware starts here.
- **Build option rot8** (7062e17): `pacman-rot8.xdc` copied onto `pacman.xdc` (tiles 1–8 on tiles 2–10), as the 2024 rot8 builds were made.

Build. With the Vivado 2020.2 settings sourced: `tcl/recreate_xpr.tcl`, then `tcl/build_xpr.tcl`.

6 Excursions

Each excursion is recorded by one `git merge -s ours` on 2425e67: the excursion's tip becomes the merge's second parent, so all its commits keep their original hashes, while the tree never changes (nothing is combined). The first-parent log is the index; a single-branch clone holds every excursion.

7 Bitstreams and their reproduction

Bitstream	Body	Commit	Vivado	Reproduction
v1.4-0.2 release	1362d6b3	0681491 (tag)	2018.3	bitwise, the tag as committed (2026-10-13); again from a fresh clone (2026-11-02)

Bitstream	Body	Commit	Vivado	Reproduction
893 (1v3 boards)	3194812d	a5f8e91 + build step	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31); reference read from a running 1v3 board
883	0ff4bf66	9fbc1dc	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31)
894 (resethack 1.8)	ec9ecc71	c98f6b0 + build step	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31); reference: the "84"/r8hack boot images
SYNC FIX	4fdfb7d2	3155d4a	2020.2	bitwise from a fresh clone with the in-repo build step (2026-11-01); reference: LBNL 1v5_20240731A_ucd_syncfix
SYNC FIX rot8	5fdf3d45	7062e17	2020.2	bitwise from a fresh clone with the in-repo build option (2026-11-01); reference: LBNL 1v5_20240810A_ucd_rot8
"74" (2x2 image)	f0b74071	—	—	not reproduced: built outside the repository, no project archive

Every bitstream above except "74" was built once more from a fresh clone of the repository with its own scripts, bitwise (finished 2026-11-02). The release tags without a released bitstream were each built once from a fresh clone as committed (Vivado 2018.3, 2026-11-03 to 05), and their bodies recorded:

Neither v1.3-0.3 nor v1.3-1.0 matches 60209db9, the bitstream of a 2022-10-31 BOOT.BIN found deleted on the 2x2 run-1 SD card, so the 1v3 line has no release reference.

8 The archive and its tags

The branch `legacy-archive-2026` (pushed 2026-11-08) is the historical record of the legacy firmware and software. Development has ended. Its first-parent history is a root commit, one merge per line, and a final commit, and every one of them holds only a `README.md`: the merges change no files, each line comes in whole as the merge's second parent with its original hashes. The README describes the merges and how to get each line back.

```
git log --first-parent legacy-archive-2026     # the merges
git branch legacy-hw1v3 f3584fe^2              # one line back (= legacy/hw1v3/883)
```

The twelve tags are annotated and named `legacy/<board>/<state>`: `legacy/hw1v3/v1.3-0.0` to `legacy/hw1v3/v1.3-1.0`, `legacy/hw1v3/893`, `legacy/hw1v3/883`, `legacy/hw1v4/v1.4-0.0`, `legacy/hw1v4/v1.4-0.2`, `legacy/hw1v4/894`, `legacy/hw1v5/syncfix` and `legacy/hw1v5/rot8`. The 893, 894 and SYNC FIX tags sit on their build steps, the commits that build. Each annotation gives the commit, the toolchain and build commands, the configuration body, the result (bitwise against its reference, or built only), the SHA-256 of the bitstream and hardware description, and the date of the build from a fresh clone. The older states (v0.0 to v1.2b-0.1) were never built and carry no tag. The larpix lightweight tags `v1.x` exist only in `larpix/pacman`.

```
git tag --merged legacy-archive-2026                    # every tag on the archive
```

Branch	Excursion	Owner	Base	Tip	What
firmware	pacman-v1.2b-1MHz	Peter Madigan	v1.2b-0.0	ba2ef0a	1 MHz MCLK for the 1v2b board
firmware	pacman-v1.3-1MHz	Peter Madigan	v1.3-0.2	38af5a7	fixed 1 MHz MCLK
firmware	pacman-v1.3-5MHz	Peter Madigan	v1.3-0.2	8a7e990	fixed 5 MHz MCLK
software	pacman-v1.2b (v1.2b-0.1)	Peter Madigan	v1.2b-0.0	6d70ed2	maintenance release (packaging)
software	pacman-1v4-petalinux20.2	Armin Karcher	v1.4-0.2	5e67360	PetaLinux 2020.2 port, eMMC boot
software	legacy-pacman-1v4-pe	Michael Mulhearn	(the port)	469c948	+ I2C fix for the MUX: the best 1v4 server

Table 2: The excursions. The fixed-MCLK lines are superseded by v1.3-1.0's adjustable MCLK.

Tag	Commit	Body
v1.3-0.0	a10a44a	17392858
v1.3-0.1	e26918e	7b70edb5
v1.3-0.2	1dfc351	7b70edb5 (no firmware change from v1.3-0.1)
v1.3-0.3	2099d0d	6f00801f
v1.3-1.0	501180f	fdda3a17
v1.4-0.0	4532a23	5aeadbf0

Table 4: Release tags built from the repository, with no released bitstream to compare with.

```
git tag -l --format='%(contents)' legacy/hw1v4/v1.4-0.2    # one tag's annotation
git checkout legacy/hw1v4/v1.4-0.2                         # the state, ready to build
```

The layout checker reads the lines from the archive's merges and also checks the archive itself: every first-parent commit holds only the README, five merges, twelve annotated tags in place. The archive and the tags are read-only on GitHub (2026-11-08): two repository rulesets forbid updates, force pushes and deletion of `legacy-archive-2026` and of every `legacy/` tag.

9 Conventions

- Historic commits keep their author, dates and hash. The snapshots, build steps, build option and excursion merges are authored by augie; the hijinks 2.X restores by Michael Mulhearn.
- Work brought in from elsewhere enters as a new commit, described by content: no cherry-pick lines, no repository-qualified branch names. Bitstream hashes and bundle names are data and may appear.
- A snapshot changes only the files its archive changes; a build step changes only build scripts and fixes without behavior change; each records its bitwise result in its message.

10 Old branches

Deleted on 2026-10-31, after a check that every commit they held is reachable from a kept branch or tag, or has its tree there: `legacy-firmware-historic`, `working-on-legacy-firmware-buildable` (the earlier design with build loops), the old excursions record, `hijinks-hw1v5-2020.2`, `legacy-pacman-1v4-pe`, `pacman-server-hijinks-2018.3`, and the tags `sync-fix` and `hijinks-2v8`. The only commit let go with a tree of its own is the first 2.8 restore (61e2fa4): the kept 2.8 plus the files its successor removes,

Merge	Line	Tip	Suggested local branch	Tags
f3584fe	hw1v3	9fbc1dc	`legacy-hw1v3`	v1.3-0.0 to 0.3, v1.3-1.0, 893, 883
df51664	hw1v4	d08d199	`legacy-hw1v4`	v1.4-0.0, v1.4-0.2, 894
c119fd5	hw1v5	7062e17	`legacy-hw1v5`	syncfix, rot8
1169b63	firmware excursions	d5fd1c7	`legacy-firmware-excursions`	
4482068	software excursions	4a27ae1	`legacy-software-excursions`	

Table 5: The merges on `legacy-archive-2026` (tip f2c89c6).

which equal 2425e67's. Six further branches (the 2020.2 GPIO-only hwcheck, the 2024 v1.5 reorganization, the 2020.2 and 2023.2 hardware demos, a 3v4 driver fixup) are under discussion.

11 Status

Done (2026-11-08). The five line branches were deleted from `mulhearn/pacman` after the deletion checks (the checker on a fresh clone, every commit reachable from `legacy-archive-2026`). "74" and 60209db9 remain unreproduced, and "74" is left to Armin to follow up if possible. The other branches still in `mulhearn/pacman` belong to hijinks and to the zipline era before `pacman-firmware` and `pacman-software`, outside this note.